

● Red Status LED

Hardware Overview

● What is it?

The **Red LED** (Light Emitting Diode) is a simple output device used to give visual feedback to the user. On your K-Comp board, this LED can indicate an error, or pulse to add atmosphere to a game.

⚡ Electrical Explanation

An LED is a specific type of **diode**, which acts like a one-way street for electricity. Current flows from the **Anode (+)** to the **Cathode (-)**, emitting light as it passes through.

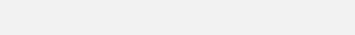
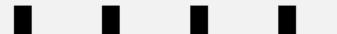
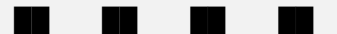
Key Hardware Facts:

- **Connection:** The `LED` is wired to **Digital Pin D5** on the microcontroller.
- **PWM Control:** Because Pin D5 supports **Pulse Width Modulation**, we can do more than just turn it **ON** or **OFF**—we can control its brightness (dimming) via software.
- **Signal Flow:** Processor (Pin D5) → Jumper (`JP_LED_L`) → Resistor → `LED` → **0V**

📡 How Dimming Works: PWM (Pulse Width Modulation)

Digital pins usually only have two states: **HIGH (5V)** or **LOW (0V)**. How do we dim the LED to 50% brightness? We cheat by switching it ON and OFF very, very fast! This technique is called **Pulse Width Modulation (PWM)**.

- The **"Duty Cycle"** is the percentage of time the signal is **ON**.
- Your eyes are too slow to see the flickering; instead, you see the *average* brightness.

Brightness	Duty Cycle	Graphical Representation
0% (OFF)	0%	 (Always Low)
25% (Dim)	25%	 (Mostly Low, short pulses)
50% (Medium)	50%	 (Half ON, Half OFF)
100% (Bright)	100%	 (Always High)

Software & Programming

📁 `kcomp_led.h`

To make controlling the `LED` easier, we use a custom library. **How to use it:** Always add `#include <kcomp_led.h>` at the very top of your code.

Function Reference

Function Name	Description & Parameters
<code>void initLed()</code>	Setup: prepares Pin D5 for output. Must be called once inside <code>setup()</code> .
<code>void ledOn(byte intensity)</code>	Turn On: sets the brightness.

	intensity is a number from 0 (OFF) to 100 (max brightness). <i>(Note: The library maps this 0–100 input to the technical 0–255 PWM value automatically).</i>
void ledOff()	Turn Off: Switches the LED off. It is the same as <code>ledOn(0)</code> .
void ledFadeIn(byte final, int duration)	Animation: smoothly brightens the LED. - final is the target brightness (0–100) - duration is time in milliseconds (e.g., 1000 = 1s).
void ledFadeOut(byte start, int duration)	Animation: Smoothly dims the LED to darkness. - start is the starting brightness (0–100) - duration is time in ms.

⚠ Pitfall: "Blocking" Code

The animation functions (`ledFadeIn` and `ledFadeOut`) are **Blocking**.

- **What this means:** When you call `ledFadeIn(100, 2000)`, the processor stays inside that function for the full 2 seconds to calculate the light curve.
- **The Consequence:** During those 2 seconds, the processor **cannot** read button presses, update the display, or check sensors.

📄 Sample Program

Below is a simple program to test your LED.

```
#include <kcomp_led.h> // 1. Include the library

void setup() {
  Serial.begin(115200);
  initLed();           // 2. Initialize the LED pin
  Serial.println("LED Ready!");
}

void loop() {
  // Turn LED on at 50% brightness
  ledOn(50);           //
  delay(1000);          // Wait 1 second

  // Smoothly fade out to 0 over 1 second (Blocking!)
  ledFadeOut(50, 1000); //

  delay(1000);          // Wait before starting again
}
```

💡 **Try it out:** You can find the full sample program **SimpleLED** in the K-Comp examples folder!